EXERCISE 2

$G=4$ treatments, $n_g = 5$ for $g = 1,\dots,4$ $\Rightarrow$ $N = \sum_{g=1}^{4} n_g = 20$

a) we have a linear model with dummy variables to encode the treatments.

Define:

$$x_{i2} = \begin{cases} 1 & \text{if tree}_i \text{ has treatment 2} \\ 0 & \text{otherwise} \end{cases} \qquad x_{i4} = \begin{cases} 1 & \text{if tree}_i \text{ has treatment 4} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{i3} = \begin{cases} 1 & \text{if tree}_i \text{ has treatment 3} \\ 0 & \text{otherwise} \end{cases}$$

then,

$$Y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \varepsilon_i \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2) \qquad i = 1,\dots,20$$

in matrix form;

observations are sorted according to the treatment:

$$\underline{Y} = [\underbrace{Y_1,\dots,Y_5}_{\text{group 1}}, \underbrace{Y_6,\dots,Y_{10}}_{\text{group 2}}, \underbrace{Y_{11},\dots,Y_{15}}_{\text{group 3}}, \underbrace{Y_{16},\dots,Y_{20}}_{\text{group 4}}]^T \qquad \underline{Y} \sim N_{20}(X\underline{\beta}, \sigma^2 I_{20})$$

the model matrix

$$X = [\underline{1} \ \underline{x}_2 \ \underline{x}_3 \ \underline{x}_4] = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} i=1 \\ \vdots \\ i=5 \\ i=6 \\ \vdots \\ i=10 \\ i=11 \\ \vdots \\ i=15 \\ i=16 \\ \vdots \\ i=20 \end{array} \qquad \text{dimension } N \times G = 20 \times 4$$

$\underline{\beta}$ is a 4-dim. vector

$$\underline{\beta} = [\beta_1 \ \beta_2 \ \beta_3 \ \beta_4]^T$$

$\underline{\varepsilon}$ is a N-dim vector

$$\underline{\varepsilon} = [\varepsilon_1 \ \dots \ \varepsilon_{20}]^T \qquad \underline{\varepsilon} \sim N_{20}(\underline{0}, \sigma^2 I_{20})$$

b) sample space $\mathcal{Y} = \mathbb{R}^{20}$

parameter space $\Theta = \mathbb{R}^4 \times (0, +\infty)$

c) in general, the sum of squares decomposition is

$$\underbrace{\sum_{g=1}^{G}\sum_{i=1}^{n_g}(y_{ig}-\bar{y})^2}_{\text{SST}} = \underbrace{\sum_{g=1}^{G}\sum_{i=1}^{n_g}(\hat{y}_{ig}-\bar{y})^2}_{\text{SSR}} + \underbrace{\sum_{g=1}^{G}\sum_{i=1}^{n_g}(y_{ig}-\hat{y}_{ig})^2}_{\text{SSE}}$$

however we know that, in this model:

$$\hat{y}_{ig} = \bar{y}_g \qquad \text{for } i = 1,\dots,n_g$$

hence

$$\sum_{g=1}^{G}\sum_{i=1}^{n_g}(\hat{y}_{ig}-\bar{y})^2 = \sum_{g=1}^{G}\sum_{i=1}^{n_g}(\bar{y}_g-\bar{y})^2 = \sum_{g=1}^{G} n_g(\bar{y}_g-\bar{y})^2 = n\sum_{g=1}^{G}(\bar{y}_g-\bar{y})^2$$

and that

$$\sum_{i=1}^{n_g}(y_{ig}-\bar{y}_g)^2 \cdot \frac{1}{(n_g-1)} = s_g^2 \Rightarrow \sum_{i=1}^{n_g}(y_{ig}-\hat{y}_{ig})^2 = \sum_{i=1}^{n_g}(y_{ig}-\bar{y}_g)^2 = (n_g-1)s_g^2 = (n-1)s_g^2$$

Hence:

$$\underset{\text{SST}}{\underset{\parallel}{\sum_{g=1}^{G}\sum_{i=1}^{n_g}(y_{ig}-\bar{y})^2}} = \underset{\text{SSR}}{\underset{\parallel}{\underbrace{n\sum_{g=1}^{G}(\bar{y}_g-\bar{y})^2}_{\substack{\text{BETWEEN-GROUP} \\ \text{SUM OF SQUARES}}}}} + \underset{\text{SSE}}{\underset{\parallel}{\underbrace{(n-1)\sum_{g=1}^{G}s_g^2}_{\substack{\text{WITHIN-GROUP} \\ \text{SUM OF SQUARES}}}}}$$

The overall mean $\bar{y} = \frac{1}{N}\sum_{g=1}^{G}\sum_{i=1}^{n_g} y_{ig} = \frac{1}{N}\sum_{g=1}^{G} n_g \bar{y}_g$

in this case $n_g = 5 = n$ for all $g = 1,\dots,G$ $\Rightarrow$ $\bar{y} = \frac{1}{N}\sum_{g=1}^{G} n \cdot \bar{y}_g = \frac{1}{Gn}\cdot n\sum_{g=1}^{G}\bar{y}_g = \frac{1}{G}\sum_{g=1}^{G}\bar{y}_g$

With the data, $\bar{y} = 0.5035$

We can compute the regression sum of squares (between-groups SS) as

$5 \cdot [(0.184 - 0.5035)^2 + (0.332 - 0.5035)^2 + (0.164 - 0.5035)^2 + (1.334 - 0.5035)^2] = 4.682 = \text{SSR}$

The error sum of squares (within-groups SS) is

$4 \cdot [0.016 + 0.319 + 0.015 + 0.737] = 4.3564 = \text{SSE}$

Finally, $\text{SST} = \text{SSR} + \text{SSE} = 9.038815$

d) the type of treatment does not have an effect $\Leftrightarrow$ the groups have the same mean weight

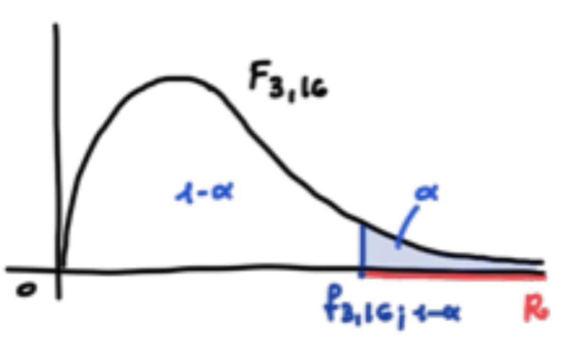
$\Leftrightarrow$ $\beta_1 = \beta_1 + \beta_2 = \beta_1 + \beta_3 = \beta_1 + \beta_4$

$\Leftrightarrow$ $\beta_2 = \beta_3 = \beta_4 = 0$

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = 0 \\ H_1: \exists r \in \{2,3,4\}: \beta_r \neq 0 \end{cases} \qquad \text{test of overall significance}$$

the test statistic is

$$F = \frac{\tilde{\Sigma}^2 - \hat{\Sigma}^2}{\hat{\Sigma}^2} \cdot \frac{N-G}{G-1} \overset{H_0}{\sim} F_{G-1,N-G} = F_{3,16}$$



and the reject region at a 5% significance level is

$$R = (f_{3,16;0.95} ; +\infty)$$
$$= (3.2389 ; +\infty)$$

the observed value of the test

$$f^{obs} = \frac{\tilde{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} \cdot \frac{16}{3}$$

where $\hat{\sigma}^2$ is the estimate of the variance of the full model (H_1)

$\tilde{\sigma}^2$ is the estimate under the null model (H_0)

$$\hat{\sigma}^2 = \frac{\text{SSE}}{N}, \qquad \tilde{\sigma}^2 = \frac{\text{SST}}{N}$$

$$\Rightarrow f^{obs} = \frac{\text{SST}-\text{SSE}}{\text{SSE}} \cdot \frac{16}{3} = \frac{\text{SSR}}{\text{SSE}} \cdot \frac{16}{3} = \frac{\text{between-groups SS}}{\text{within-groups SS}} \cdot \frac{16}{3} = \frac{4.682}{4.3564} \cdot \frac{16}{3} = 5.7325$$

$f^{obs} \in R$ $\Rightarrow$ reject H_0 at a 5% significance level.

e) We know that, according to the specified model

$$\begin{cases} \mu_1 = \beta_1 \\ \mu_2 = \beta_1 + \beta_2 \\ \mu_3 = \beta_1 + \beta_3 \\ \mu_4 = \beta_1 + \beta_4 \end{cases} \Leftrightarrow \begin{cases} \beta_1 = \mu_1 \\ \beta_2 = \mu_2 - \mu_1 \\ \beta_3 = \mu_3 - \mu_1 \\ \beta_4 = \mu_4 - \mu_1 \end{cases}$$

Moreover, the MLE of μ_g is $\bar{y}_g = \frac{1}{n}\sum_{i=1}^{n} y_{ig}$ $\qquad g = 1,\dots,G$

Hence, the MLE of $\underline{\beta}$ is

$$\hat{\underline{\beta}} = \begin{bmatrix} \bar{y}_1 \\ \bar{y}_2 - \bar{y}_1 \\ \bar{y}_3 - \bar{y}_1 \\ \bar{y}_4 - \bar{y}_1 \end{bmatrix} = \begin{bmatrix} 0.184 \\ 0.148 \\ -0.02 \\ 1.15 \end{bmatrix}$$

The estimated model is $\hat{y}_i = 0.184 + 0.148\, x_{i1} - 0.02\, x_{i2} + 1.15\, x_{i3}$

f) $\hat{\underline{B}} \sim N_4(\underline{\beta}, (X^TX)^{-1}\sigma^2)$

and

$\hat{B}_2 \sim N(\beta_2, 0.40\,\sigma^2)$

g) $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$

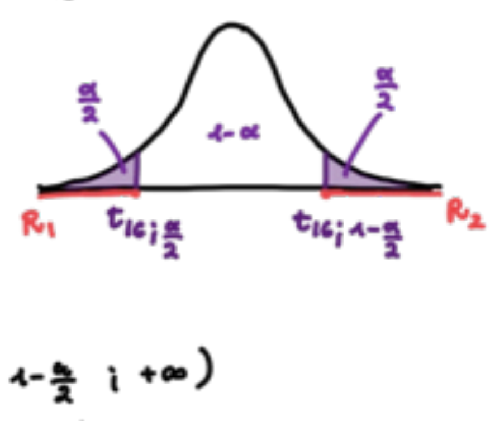
the distribution of $\hat{B}_2$ is given above, but σ^2 is unknown

$$\frac{\hat{B}_2}{\sqrt{0.40\,\sigma^2}} \overset{H_0}{\sim} N(0,1)$$

We estimate σ^2:

$$\frac{\hat{B}_2}{\sqrt{0.40\, S^2}} \overset{H_0}{\sim} t_{N-G} = t_{16}$$

Two-tail reject region



$$R = R_1 \cup R_2$$
$$= (-\infty ; -t_{16;1-\frac{\alpha}{2}}) \cup (t_{16;1-\frac{\alpha}{2}} ; +\infty)$$
$$= (-\infty ; -2.119) \cup (2.119 ; +\infty)$$

$\frac{\alpha}{2} = 0.025$

$1-\frac{\alpha}{2} = 0.975$

The observed value of the test is:

$$t^{obs} = \frac{\hat{\beta}_2}{\sqrt{0.40\, s^2}} = \frac{0.148}{\sqrt{0.40 \cdot \frac{4.3564}{16}}} = 0.44$$

$t^{obs} \notin R$ | do not reject H_0

Hence, treatments (1) and (2) are not statistically different

Applying no treatment or only fertilizer have the same effect on the trees' weight.

h) $R^2 = \frac{\text{SSR}}{\text{SST}} = 0.5180$

It represents the proportion of variability of Y that is explained by the model.

Here, we explain about half of the total variability.

i) $\sum_{i=1}^{20} e_i = 0$ $\qquad$ yes, because the model includes the intercept.

$\Rightarrow$ $\underline{1} \in \mathcal{C}(X)$ and we know that $\underline{e}^T\underline{v} = 0$ for all $\underline{v} \in \mathcal{C}(X)$

$\Rightarrow$ $\underline{e}^T\underline{1} = \sum_{i=1}^{20} e_i = 0$

$$\sum_{i=1}^{5} e_i = 0 \qquad \text{yes}: \sum_{i=1}^{5} e_i = \sum_{i=1}^{20} e_i - \sum_{i=6}^{20} e_i = \underline{e}^T\underline{1} - (\underline{e}^T(\underline{x}_2 + \underline{x}_3 + \underline{x}_4))$$
$$= \underline{e}^T \underbrace{(\underline{1} - \underline{x}_2 - \underline{x}_3 - \underline{x}_4)}_{\substack{\text{linear combination} \\ \text{of vectors } \in \mathcal{C}(X) \Rightarrow \in \mathcal{C}(X)}} = 0$$

$$\sum_{i=1}^{10} e_i = 0 \qquad \text{yes} \qquad \sum_{i=1}^{10} e_i = \underline{e}^T(\underline{1} - \underline{x}_3 - \underline{x}_4) = 0$$

$$\sum_{i=4}^{12} e_i = 0 \qquad \text{no}$$